PRABHU KUMAR
Technical Specialist, Machine Learning
prabhundps@gmail.com | +91-8521987766/+91-7619117901 | Bengaluru, India
View GitHub Profile | View LinkedIn Profile

"Almost 8+ years building Machine Vision AI end-to-end pipeline - data, training, optimization, and hardware deployment. Currently on autonomous vehicle perception. Strong math foundation. Seeking impactful CV/ML/Gen AI roles. Open to relocate/Remote."

Professional Experience

Technical Specialist - Machine Learning, Continental Automotive Pvt. Ltd, Bengaluru, India, Oct'18-Present

- Developing end-to-end model deployment pipeline of AI-based anomaly detector on Ambarella SoC: analyzed raw sensor data from IMX728/IMX623 camera sensor, injected synthetic EE faults using encoder-decoder based AI model for ASIL-D safety compliance, researched and implemented optimal deep learning architectures, trained models on large-scale image datasets in AI clusters, optimized for 2ms latency and 3 eTOPS, and successfully deployed on AMBARELLA CV3-AD 685 board.
- Designed and optimized advanced **deep learning models** for object detection & tracking, trajectory prediction, path planning, sensor fusion and other multiple POCs on ADAS applications, and achieving up to 98% accuracy on inhouse camera sensor dataset.
- Converted models to **ONNX** format for seamless hardware integration, and leading to a 20–30ms reduction in **deployment latency [**3ms-5ms**]**.
- Built end-to-end **Computer/Machine Vision applications** for synthetic data generation of **Camera Blockage** and **Camera Safety Monitor** components and established **data pipelines for model training and validation.**
- **Generated** foggy weather **training datasets** leveraging **depth estimation** techniques and incorporating shadow removal in complex scenarios to enhance **data quality** for adverse weather conditions.
- Developed a PointNet-based deep learning architecture for **classification of radar point-cloud data**, and achieving performance metrics of up to 97%, outperforming camera-based detection systems in identifying traffic participants.
- Forecasted **agent vehicle trajectories** in complex traffic scenarios by integrating HD maps and environmental context using **Graph Neural Networks (GNNs)** with open-source datasets to enhance prediction accuracy.
- Unified **multi-sensor data fusion** and **tracking** by synchronizing inputs from MFC cameras and ARS/SRR radar sensors, dramatically reducing uncertainty. Implemented a joint **mid-level feature extraction framework** redefining multi-object and lane tracking performance.
- **Leading** a technical team of size 5.

Research Intern, E.I. DuPont Pvt. Ltd, Hyderabad, India, Sep'18

- Conceived, engineered**,** and optimized deep learning architecture for drone-based image segmentation on customized 5K ground truth images with IoT device configuration for Proof of concept with tangible outcome.
- **Plant classification and segmentation system-** Brainstormed computer vision techniques, including locally adaptive histogram equalization to prepare ground truth and train a deep learning segmentation model. OpenCV.

Data Science Intern, Cyrrup Solutions, India, Feb'18

- Managed and drafted complete project cycle: data collection, data preparation, data analysis, and neural network training. Achieved notable 95% accuracy in drowsiness detection and nearly 97% accuracy in driver behavior classification, deployed on Raspberry Pi.

Technical Proficiencies

- **Computer Vision and Machine Vision**: Image Processing, Segmentation, Classification, Object detection, Multi Object Tracking and Data association, Sensor Fusion, Image Restoration, Image upscaling, Anomaly Detection, Camera sensor's Signal Processing Flow with RAW image pipeline
- **Deep Learning and Advanced AI:** Convolution Neural Network, Transformer, Large Language Models, Vision Transformer, Generative Adversarial Network/ GANs, LLMs, VLMs, Stable Diffusion, Multi-Model AI, Mix of Experts, Reasoning Model, and Reinforcement Learning
- **Applied ML for Production:** Transfer learning, Domain adaptation, Neural Architecture search, Agentic AI, and Recommender System
- **Efficient AI and Model Compression**: Knowledge Distillation, Pruning, Quantization, Efficient ML training, and Efficient ML Inference, and Edge AI
- **Deep-Learning Frameworks**: PyTorch, PyTorch-lightning, Keras, TensorFlow, and ONNX
- **GPU and Distributed Training**: Parallel Processing, Parallelism, Distributed Neural Network training and performance optimization on GPUs,
- Data Foundations: Probability, Statistics, Linear Algebra, Calculus, Graph Theory, and Data Wrangling and Data Engineering for AI Pipelines
- **Languages**: Python, C, C++

Professional Developments

- CME 295: Transformer & Large Language Model, Oct'25
- MIT 6.5940: TinyML and Efficient Deep Learning Computing, June'25
- CS336: Language Models from Scratch, Dec' 2025
- Organized a "Hackathon'23, Technical Center India, Continental", Sep'23
- Professional certificate: TensorFlow Developer, Deeplearning.ai, April'22.
- Digital Image Processing, NPTEL, Oct'22.
- Social and Economic Networks: Model and Analysis, Coursera, June'20.
- Data Science and Informatics Hackathon, April '18.
- PyImageSearch Gurus, Deep Learning and OpenCV in Python Dec'17

Education

- MicroMaster's Program, Statistics and Data Science, Massachusetts Institute of Technology, Jan'22 – Jan'24.
- M. Tech, Computer Science & Engineering, IIT (ISM) Dhanbad, Jharkhand, July'15- June'17, CGPA- 7.54/10.0

Achievements and awards

- Recognized for "Key player of the year, 2022-2023" Continental Key Player Award, 2023
- Earned Spot Award, Continental Automotive Pvt. Ltd, 2020 & 2022.
- Awarded for "Outstanding performance in Jharkhand State" at Higher Secondary School Certification Exam, Mar'11